

MACHINE LEARNING UNIVERSITY

Machine Learning University - Educator Enablement Program

This hands-on activity will explore Amazon's Agentic AI workspace: **Amazon Quick**. Quick will guide you to create an app as described below.

What we are building

Study Plan Generator — Features & Requirements

Overview

A web app that allows users to upload a slide deck, set a due date, and generate a research-backed study plan powered by AI and web search.

Features

1. File Upload & Parsing

- **Drag-and-drop** or click-to-browse file upload interface
- **Supported formats:** PDF, PPTX, TXT, MD, CSV, HTML/HTM
- Displays upload status with a "remove" option to reset
- Visual drag-over feedback

2. Topic Extraction (AI)

- Uses **Claude Sonnet** model with structured tool output
- Extracts **3–8 distinct topics** from the first 8,000 characters of the slide content
- Employs **defineTool()** for guaranteed structured JSON output

3. Academic Paper Search (Web Search)

- For each extracted topic, performs a web search using the **Quick Suite Web Search** connector (quicksuite-websearch)
- Queries are appended with "research papers academic" for academic relevance
- Returns up to **6 citations per topic** with title, URL, snippet, and domain
- **Sanitizes** search queries by removing control characters and limiting to 200 chars
- Gracefully handles search failures per topic (shows "No papers found")

4. Study Plan Generation (AI)

- Uses **Claude Sonnet** to produce:

- **Code-Free Teaching Approach** — tailored for social sciences and humanities
- **Code-Based Teaching Approach** — tailored for engineering, CS, and sciences
- **Week-by-week Timeline** — structured schedule to cover all topics before the due date
- Accepts an optional due date input

5. Results Browser

- **Topic filter** dropdown — view all topics or one at a time
- **Persona filter** — toggle between All / Code-Free / Code-Based views
- **Collapsible study plan panel** (Show/Hide toggle) with:
 - Timeline display
 - Teaching approaches per persona
- **Citation cards** — clickable links to external papers with title, domain, and snippet preview

Technical Requirements & Constraints

Area	Requirement
File size	Input truncated to 8,000 chars for AI processing; output token limits of 1024/2048

Outstanding / Potential Enhancements

- **Refactoring** — *StudyPlanGenerator.tsx* (~220 lines) could be split into smaller modules (topic extractor, search service, plan generator)
- **PDF export** — Export the generated study plan as a downloadable PDF
- **Slide content preview** — Collapsible section showing extracted raw text
- **Progress indicator** — Per-topic status checklist during generation (instead of single status line)

Before You Start

Make sure you have:

- An Amazon Quick account (ask your instructor or admin for access)
- A web browser (Chrome, Edge, or Firefox)
- Some study materials ready to upload (syllabus, lecture notes, readings such as PDF or Word files)

Step 1: Open the Amazon Quick Web App

1. Open your browser and go to your Amazon Quick web app URL (<https://quick.aws.com/sn>).
2. Click the Sign in button
3. Log in with your account credentials.
4. You'll land on the **Amazon Quick homepage**.

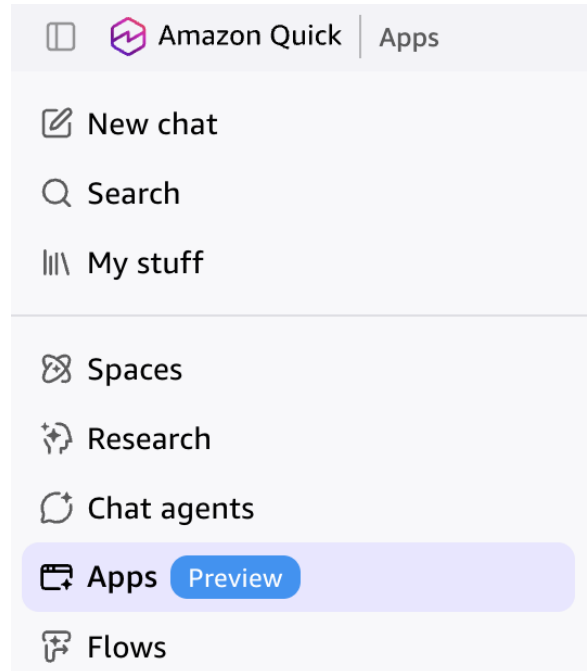
Tip: If you've been using the browser extension, click **"Go to web app"** from the dropdown menu at the top of the side panel.

Create a New App

It is time to guide Amazon Quick to create your app according to the features described above.

Ask Quick to create the app

1. Click the **Apps** option on the left menu

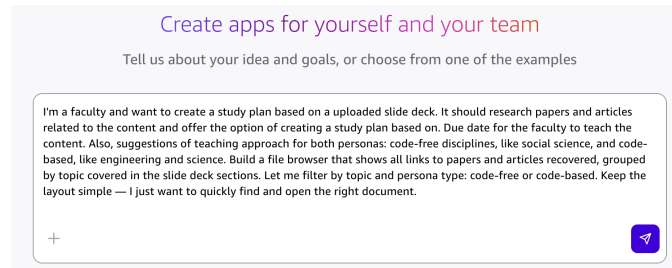


Screenshot: Quick Apps

2. Copy the prompt below, which describes the features Quick needs to build for your app:

"Build a single-page React web application called "Faculty Study Plan Generator" that lets faculties upload a slide deck, set a due date, and generate a research-backed study plan using AI and web search. The app runs inside a sandboxed iframe (allow-scripts, allow-modals) with no external dependencies beyond pre-installed packages (pdf-lib, React, runtime-lib).FEATURE 1: FILE UPLOAD & PARSING - Implement a drag-and-drop (with click-to-browse fallback) file upload interface. Support these formats: PDF, PPTX, TXT, MD, CSV, HTML/HTM. Show upload status with a "remove" button to reset. Provide visual drag-over feedback (e.g., border highlight). FEATURE 2: TOPIC EXTRACTION - Use Claude Haiku as model for topic extraction. From the first 8,000 characters of the uploaded file content, extract 3–8 distinct topics. Output token limit: 1024. FEATURE 3: ACADEMIC PAPER SEARCH - For each extracted topic, perform a web search using the Quick Suite Web Search connector. Append "research papers academic" to each query for academic relevance. Return up to 6 citations per topic, each with: title, URL, snippet, and domain. Handle search failures gracefully per topic — display "No papers found" instead of crashing. FEATURE 4: STUDY PLAN GENERATION - Use Claude Haiku to generate a study plan with three sections: 1. Code-Free Teaching Approach — tailored for social sciences and humanities learners. 2. Code-Based Teaching Approach — tailored for engineering, CS, and sciences learners. 3. Week-by-week Timeline — a structured schedule to cover all topics before the user-specified due date. Accept an optional due date input (date picker). Output token limit: 2048.FEATURE 5: RESULTS BROWSER - Topic filter dropdown — view all topics or select one at a time. Persona filter — toggle between All / Code-Free / Code-Based views. Collapsible study plan panel with a Show/Hide toggle containing: 1. The week-by-week timeline, 2. Teaching approaches per persona. Citation cards — clickable links to external papers showing title, domain, and snippet preview."

3. Now, paste the prompt into the prompt window in the **Apps** page like below:

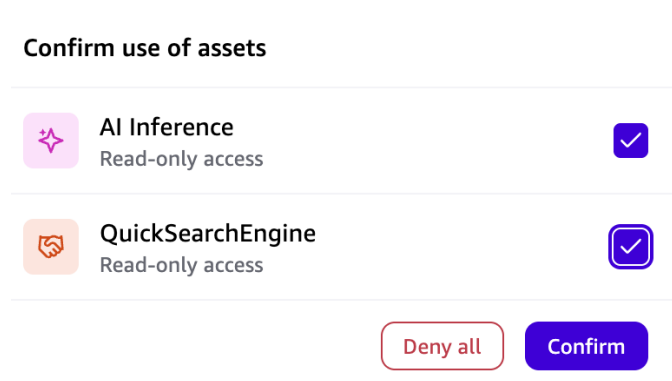


Screenshot: app creation prompt

Follow what Quick is doing

Quick will start building your app. Keep looking at the messages. Note how, sometimes, Quick runs into an error and solve it autonomously.

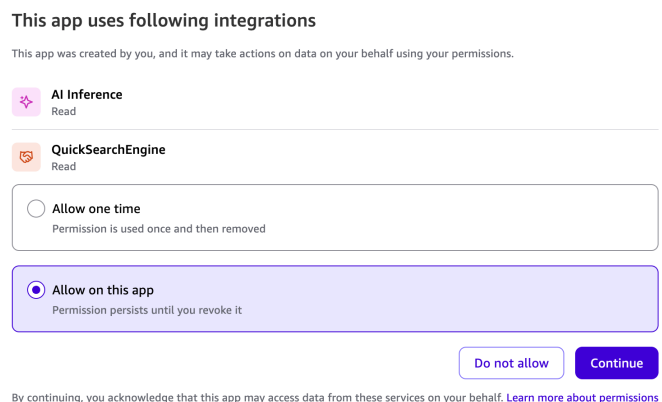
Quick may ask for your action, like in the case below:



Screenshot: Ask for user action

Mark both and click **Confirm**.

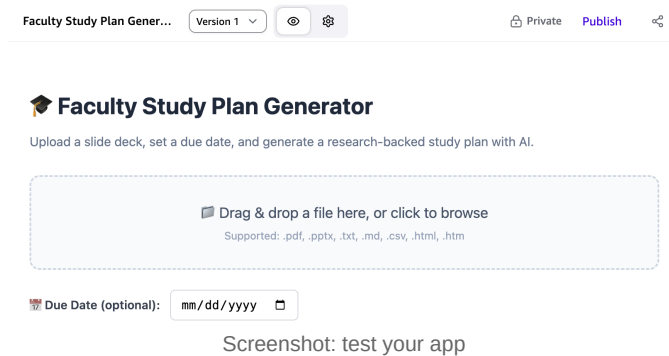
When the app is ready, Quick needs to ask you how to handle the integrations with the available tools. In this case, select the option **Allow on this app** and click the **Continue** button.



Screenshot: Continue button

Testing your app

Now that your app is ready, Quick will offer the app interface. It should look like the image below:



Time to test your app. Run your app and keep asking Quick to improve it.

Notice, at the top of the previous image, how Quick has created the **Version 1** of your app.

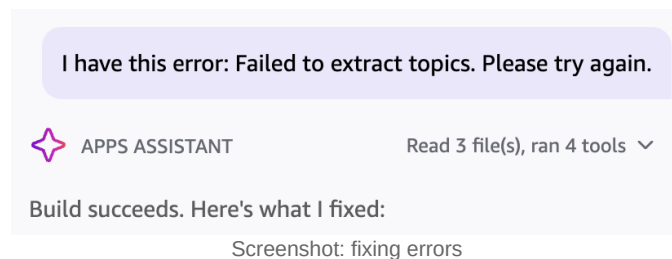
1. Start by uploading the slide deck from one of the modules in the AWS-EEP Agentic AI content:
 - Lesson - Agentic AI Essentials
 - Lesson - Agentic AI Applications
2. Optionally, select a due date and click in the button to generate your study plan

Fixing your app

Benjamin Franklin famously said only two things in life are certain: death and taxes. I'd like to propose a third — your AI assistant spectacularly face-planting mid-task.

So here's the deal: building apps with AI is a bit like working with a very enthusiastic intern who occasionally sets the kitchen on fire. It will break things. Your job is to patiently point at the flames, say "no, not that," and keep nudging it until the app actually works. Persistence beats perfection.

If something is not working as expected or you have an error message, share the message in the chat window and ask Quick to fix it.

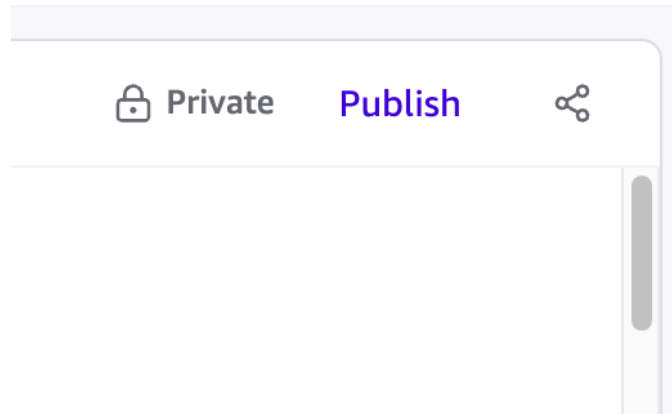


It will update your app, creating new versions, allowing you to come back to a previous version if you are not happy with any new version created.

Share your App!

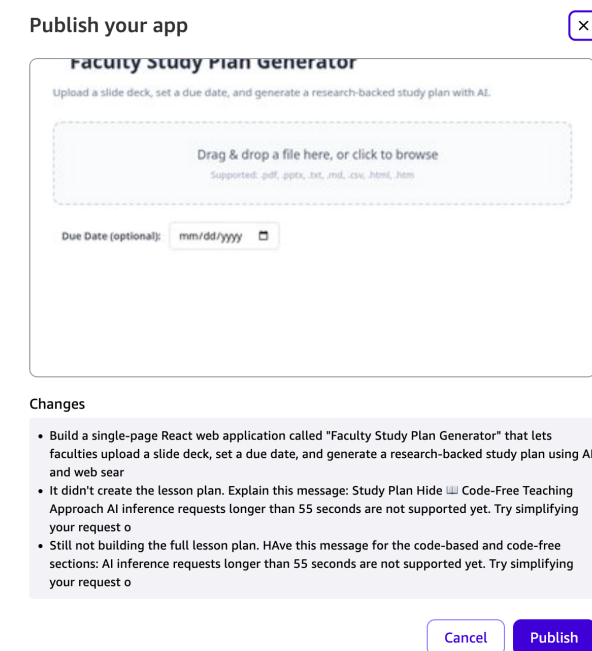
Once you app is running properly, you can share it with your peers.

1. Just click in the **Publish** button at the top right of your app's page



Screenshot: Publish button

2. Click **Publish** again



Screenshot: Confirm publish

3. Copy the link and share with your peers!

Your app has been published



Faculty Study Plan Generator

Upload a slide deck, set a due date, and generate a research-backed study plan with AI.

Drag & drop a file here, or click to browse

Supported: .pdf, .pptx, .txt, .md, .csv, .html, .htm

Due Date (optional):

App name: Faculty Study Plan Generator

URL:

<https://us-east-1.quicksight.aws.amazon.com/sn/account/amazonbi/apps/>



Dismiss

Open

Screenshot: [Copy link](#)

Congrats!

You have created a app without needing to write a single line of code!

Under the hood, Amazon Quick has created a complex app based on your prompt!

Keep using Amazon Quick and have fun!